

**WORKED KEY**

Full Name: \_\_\_\_\_

Period: \_\_\_\_\_

**This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.**

1. Solve the system by graphing.  $y = x + 4$   
 $y = -2x + 1$

**Answer:**  $(-1, 3)$

2. Graph the system. What does the graph tell you about its solutions?  $y = -2x + 5$   
 $4x + 2y = 10$

**Answer:** infinitely many solutions — the second equation is 2× the first: same line

3. Solve using substitution:  $y = x - 4$   
 $3x + y = 16$

**Answer:**  $3x + (x - 4) = 16 \rightarrow 4x = 20 \rightarrow x = 5, y = 1 \rightarrow (5, 1)$

4. Solve using substitution:  $x = 4y + 1$   
 $2x + 3y = 24$

**Answer:**  $2(4y + 1) + 3y = 24 \rightarrow 11y = 22 \rightarrow y = 2, x = 9 \rightarrow (9, 2)$

5. Solve using elimination:  $2x + 3y = 22$   
 $3x - 3y = 3$

**Answer:** add:  $5x = 25 \rightarrow x = 5; 2(5) + 3y = 22 \rightarrow y = 4 \rightarrow (5, 4)$

6. Solve using elimination:  $3x + y = 11$   
 $4x + 3y = 23$

**Answer:** multiply the first by 3:  $9x + 3y = 33$ ; subtract:  $5x = 10 \rightarrow x = 2, y = 5 \rightarrow (2, 5)$

# Algebra 1 – Module 7 Practice Test •

## WORKED KEY

7. Without graphing, decide how many solutions the system has. Justify.  $4x - 2y = 6$      $2x - y = 5$

**Answer:** doubling the second gives  $4x - 2y = 10 \neq 6$  — parallel lines: no solution

8. Solve using any method:  $y = 7x + 1$   
 $y = 3x + 13$

**Answer:**  $7x + 1 = 3x + 13 \rightarrow 4x = 12 \rightarrow x = 3, y = 22 \rightarrow (3, 22)$

9. Carnations cost \$3 and roses cost \$5. The student council sells 20 flowers for \$76. Write and solve a system to find how many of each were sold.

**Answer:**  $c + r = 20, 3c + 5r = 76 \rightarrow 3(20 - r) + 5r = 76 \rightarrow 60 - 3r + 5r = 76 \rightarrow 2r = 16 \rightarrow r = 8$  roses,  $c = 12$  carnations

13. A rectangular flag's length is 5 in. more than twice its width, and its perimeter is 52 in. Write and solve a system to find the dimensions.

**Answer:**  $l = 2w + 5, 2l + 2w = 52 \rightarrow 2(2w + 5) + 2w = 52 \rightarrow 6w = 42 \rightarrow w = 7$  in.,  $l = 19$  in.

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10. Four candles and three vases cost \$37.00. Four candles and one vase cost \$23.00. Find the price of each item.

**Answer:** subtract:  $2v = 14 \rightarrow$  vase \$7.00;  $4c + 7 = 23 \rightarrow 4c = 16 \rightarrow$  candle \$4.00

11. Error Analysis — Solving  $x + 2y = 10$   
 $3x + 5y = 28$ , Rex added the equations right away and got  $4x + 7y = 38$ , which still has both variables. What did he forget, and what is the solution?

**Answer:** adding only works when a variable has OPPOSITE coefficients — multiply the first by  $-3$  first:  $-3x - 6y = -30$ ; add:  $-y = -2 \rightarrow y = 2, x = 6$

12. Studio A charges a \$50 registration fee plus \$15 per lesson; Studio B charges a \$20 registration fee plus \$20 per lesson. After how many lessons do they cost the same, and what is that cost?

**Answer:**  $50 + 15n = 20 + 20n \rightarrow 30 = 5n \rightarrow n = 6$  lessons; cost \$140

## Cumulative Review — ACT Style. Circle the best answer.

14. Solve:  $7 - 2x < 1$

(A)  $x > 3$   
(B)  $x < 3$   
(C)  $x > -3$   
(D)  $x < -3$

**Answer:**  $-2x < -6 \rightarrow$  FLIP:  $x > 3$

15. What slope is perpendicular to  $y = (2/3)x - 4$ ?

(E)  $2/3$   
(F)  $-2/3$   
(G)  $3/2$   
(H)  $-3/2$

**Answer:** opposite reciprocal of  $2/3$

16. What is the slope of the line  $10x - 2y = 8$ ?

(A) 10  
(B) 5  
(C)  $-5$   
(D)  $-2$

**Answer:**  $-2y = -10x + 8 \rightarrow y = 5x - 4$